

Organic Chemistry Practical: Synoptic Organic Deductions & Structural Elucidation

10 Most Frequently Asked Practical Examination Questions (Past 4 Years Analysis: 2021–2024)

CANDIDATE NAME	URWAH	CENTRE NUMBER	PK082
COMPONENT	Paper 3 — Advanced Practical Skills (Past 10 Years Archive)	CANDIDATE NUMBER	0142
TIME ALLOWED	2 Hours 30 Minutes	TOTAL MARKS	100 Marks

INSTRUCTIONS TO CANDIDATES:

- Answer all questions in the spaces provided on the question paper.
- Give working for all calculations and record all experimental readings to the required precision:
 - Burette readings to the nearest 0.05 cm³; balance readings to 0.01 g / 0.001 g.
 - Thermometer readings to the nearest 0.5 °C.
- You may use an HB pencil for any rough working, graphs, or apparatus diagrams.
- Non-exact numerical answers should be given to three significant figures unless stated otherwise.
- You will need a scientific calculator and the Qualitative Analysis Notes provided on pages at the back.

LABORATORY INVESTIGATION OVERVIEW:

This laboratory pack contains authentic Cambridge Advanced Practical examinations. Each practical scenario provides complete procedures, chemical specifications (FA codes), hazard warnings, structured data collection tables, graphical plotting grids, and comprehensive multi-step mathematical and observational deductions.

Question 1: Deductive Elucidation of the Four Structural Isomers of C₄H₁₀O**Cambridge Syllabus Reference: 9701/31/M/J/23/Q3 · [10 Marks]**

Four liquids, W, X, Y, and Z, are the four constitutional isomers of C₄H₁₀O.

- W, X, and Y react with acidified potassium dichromate(VI) to form green solutions; Z does not react.
- Only X gives a yellow precipitate with alkaline aqueous iodine.
- W and Y are oxidized to aldehydes; W has a higher boiling point than Y.

(a) Identify isomer Z and explain why it does not react with acidified dichromate. [2]

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(b) Identify isomer X and state the formula of the yellow precipitate. [2]

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(c) Identify isomers W and Y, explaining the difference in their boiling points. [4]

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(d) Draw the skeletal formula of isomer X. [2]

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Question 2: Deductive Identification of an Unknown Bifunctional Organic Acid (C₄H₆O₂)

Cambridge Syllabus Reference: 9701/32/M/J/22/Q3 · [10 Marks]

An unknown compound, Acid P (formula C₄H₆O₂), exhibits the following properties:

- Decolourizes bromine water instantly from orange to colorless.
- Gives vigorous effervescence with solid NaHCO₃.
- Exists as a pair of diastereomers (geometric isomers).

(a) Identify the two functional groups present in Acid P. [2]

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(b) Draw the displayed formula and state the systematic IUPAC name of Acid P. [3]

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(c) Explain why Acid P exhibits geometric (cis/trans or E/Z) isomerism. [3]

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(d) Draw and label the E and Z isomers of Acid P. [2]

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Question 3: Differentiation of Carbonyl Constitutional Isomers of Molecular Formula C₄H₈O

Cambridge Syllabus Reference: 9701/34/O/N/23/Q3 · [10 Marks]

Three compounds, J, K, and L, have molecular formula C₄H₈O.

- All three compounds react with 2,4-DNPH to form orange precipitates.
- Compounds J and K reduce Tollens' reagent; compound L does not.
- Compound L gives a yellow precipitate with alkaline aqueous iodine.
- Compound J has an unbranched carbon chain; compound K has a branched chain.

(a) Classify J, K, and L as aldehydes or ketones. [2]

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(b) Deduce the structural formula and systematic name of compound L. [3]

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(c) Deduce the structural formula and systematic name of compound J. [2]

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(d) Deduce the structural formula and systematic name of compound K. [3]

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Question 4: Geometric Stereoisomerism in Unsaturated Dicarboxylic Acids: Maleic vs Fumaric Acid

Cambridge Syllabus Reference: 9701/33/M/J/24/Q3 · [10 Marks]

Compound M and Compound N are geometric isomers of butenedioic acid, $\text{HOOC}-\text{CH}=\text{CH}-\text{COOH}$.

- Compound M (cis / Z) dehydrates readily at 130°C to form a cyclic anhydride (maleic anhydride).
- Compound N (trans / E) does not form a cyclic anhydride at 130°C , requiring heating above 250°C .

(a) Draw the structural formula of Compound M and Compound N clearly showing the geometry. [4]

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(b) Explain in terms of molecular geometry why Compound M easily forms a cyclic anhydride while Compound N does not. [4]

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(c) State whether maleic acid or fumaric acid has a higher melting point. Explain. [2]

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Question 5: Identification and 3D Representation of Chiral Centres in Practical Unknowns

Cambridge Syllabus Reference: 9701/35/M/J/23/Q3 · [10 Marks]

A student hydrolyzes 2-bromobutane with aqueous sodium hydroxide. The organic product is isolated and analyzed for optical activity.

(a) Identify the organic product formed and state whether it possesses a chiral centre. [2]

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(b) Identify the four different groups attached to the chiral carbon atom in this molecule. [2]

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(c) Draw 3D wedge-and-dash representations of the two non-superimposable mirror-image enantiomers. [4]

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(d) Explain how the two enantiomers behave when exposed to plane-polarized light. [2]

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Question 6: Classification of Alcohols using Lucas Reagent (Conc. HCl and Anhydrous ZnCl₂)

Cambridge Syllabus Reference: 9701/31/O/N/22/Q3 · [10 Marks]

Lucas reagent is a solution of anhydrous zinc chloride in concentrated hydrochloric acid. It reacts with alcohols via nucleophilic substitution to form insoluble liquid chloroalkanes, causing cloudiness.

- (a) Describe the observations and reaction times when Lucas reagent is added at room temperature to: (i) 1° alcohol, (ii) 2° alcohol, (iii) 3° alcohol. [4]

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- (b) Identify the chemical substance causing the cloudiness and phase separation. [2]

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- (c) Explain the extreme difference in reaction rates in terms of carbocation stability. [4]

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Question 7: Deductive Identification of an Unknown Halogenoalkane by Hydrolysis Rate and Density

Cambridge Syllabus Reference: 9701/32/M/J/24/Q3 · [10 Marks]

An unknown halogenoalkane, Liquid Y, has molecular formula C_3H_7X .

- When shaken with water, Liquid Y sinks to the bottom (density $> 1.0 \text{ g cm}^{-3}$).
- When warmed with aqueous NaOH, acidified with HNO_3 , and tested with $AgNO_3$, a cream precipitate forms within 2 minutes.
- The cream precipitate is insoluble in dilute NH_3 but dissolves in concentrated NH_3 .

(a) Identify the halogen atom present in Liquid Y from the silver nitrate test. [2]

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(b) Liquid Y exists as two constitutional isomers. Draw and name both isomers. [4]

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(c) When hydrolysed, the resulting alcohol does NOT give a yellow precipitate with alkaline aqueous iodine. Deduce the exact structure of Liquid Y. [2]

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(d) Write the ionic equation for the precipitation of silver bromide. [2]

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Question 8: Diagnostic Deductions of Nitrogen-Containing Organic Compounds: Amines vs Amides

Cambridge Syllabus Reference: 9701/34/M/J/22/Q3 · [10 Marks]

Compound Q is an organic liquid containing nitrogen.

- When tested with universal indicator paper, Q turns paper blue-green (pH ~ 9).
- When glass rod dipped in concentrated HCl is held near open bottle of Q, dense white fumes form.
- When Q is mixed with ethanoyl chloride, vigorous reaction occurs forming misty acidic fumes and a white solid.

(a) Identify the functional group class of Compound Q. [2]

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(b) Identify the chemical formula of the white fumes formed with concentrated HCl. [2]

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(c) Explain why amines act as Bronsted-Lowry bases in aqueous solution. [3]

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(d) Write the balanced chemical equation for the reaction of ethylamine with ethanoyl chloride. [3]

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Question 9: Deductive Structure Determination Combining Elemental Analysis and Titrimetric Mr

Cambridge Syllabus Reference: 9701/35/O/N/22/Q3 · [10 Marks]

A solid organic acid, Compound R, contains 40.0% C, 6.7% H, and 53.3% O by mass.
A 0.900 g sample of Compound R is dissolved in water and titrated with 0.200 mol dm⁻³ NaOH. Exactly 50.00 cm³ of NaOH is required to reach the phenolphthalein endpoint.

(a) Calculate the empirical formula of Compound R. [Ar: C = 12.0, H = 1.0, O = 16.0] [3]

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(b) Calculate the moles of NaOH reacted and determine the relative molecular mass (Mr) of Compound R, assuming it is a monoprotic acid. [3]

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(c) Determine the molecular formula of Compound R. [2]

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(d) Compound R gives a positive tri-iodomethane (iodoform) test. Deduce the structural formula and systematic name of Compound R. [2]

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Question 10: Comprehensive Laboratory Decision Tree for the Identification of Unknown Organic Liquids

Cambridge Syllabus Reference: 9701/33/O/N/23/Q3 · [10 Marks]

Candidates in AS Chemistry practical exams are presented with unknown liquids and must follow a logical, systematic deductive sequence.

(a) State the first test performed to determine if an unknown liquid is a carbonyl compound. [2]

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(b) If the 2,4-DNPH test is positive, state the definitive test used to distinguish whether it is an aldehyde or ketone. [2]

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(c) If the 2,4-DNPH test is negative, state the test used to check for a carboxylic acid versus an alcohol. [3]

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(d) If an alcohol is confirmed, state how to differentiate primary/secondary from tertiary alcohols. [3]

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Examiner Mark Scheme & Indicative Observation Guide

Confidential Marking Points & Error Tolerances for Organic Chemistry Practical: Synoptic Organic Deductions & Structural Elucidation

Question 1: Deductive Elucidation of the Four Structural Isomers of C₄H₁₀O [10 Marks]

Part	Marking Points & Expected Observations	Marks
(a)	Z is 2-methylpropan-2-ol ((CH ₃) ₃ COH) [1]; it is a tertiary alcohol with no alpha-hydrogen atom [1].	[2]
(b)	X is butan-2-ol (CH ₃ CH ₂ CH(OH)CH ₃) [1]; precipitate is CHI ₃ [1].	[2]
(c)	W is butan-1-ol (CH ₃ CH ₂ CH ₂ CH ₂ OH); Y is 2-methylpropan-1-ol ((CH ₃) ₂ CHCH ₂ OH) [2]; W is unbranched and has greater molecular surface area of contact, resulting in stronger London dispersion forces than branched Y [2].	[4]
(d)	Correct 4-carbon chain with -OH group on C2 [2].	[2]

Question 2: Deductive Identification of an Unknown Bifunctional Organic Acid (C₄H₆O₂) [10 Marks]

Part	Marking Points & Expected Observations	Marks
(a)	Alkene (C=C double bond) [1]; Carboxylic acid (-COOH) [1].	[2]
(b)	Formula: CH ₃ -CH=CH-COOH showing all bonds [1.5]; Name: But-2-enoic acid (or 2-butenic acid) [1.5].	[3]
(c)	Restricted rotation around the carbon-carbon double bond (C=C) due to pi-bond overlap [1.5]; each carbon atom of the C=C double bond is bonded to two different groups (-H and -CH ₃ on one C; -H and -COOH on the other) [1.5].	[3]
(d)	E-isomer with -CH ₃ and -COOH on opposite sides [1]; Z-isomer with -CH ₃ and -COOH on same side [1].	[2]

Question 3: Differentiation of Carbonyl Constitutional Isomers of Molecular Formula C₄H₈O [10 Marks]

Part	Marking Points & Expected Observations	Marks
(a)	J and K are aldehydes (reduce Tollens') [1]; L is a ketone (fails Tollens') [1].	[2]
(b)	Formula: CH ₃ COCH ₂ CH ₃ [1.5]; Name: Butanone (or butan-2-one) [1.5].	[3]
(c)	Formula: CH ₃ CH ₂ CH ₂ CHO [1]; Name: Butanal [1].	[2]
(d)	Formula: (CH ₃) ₂ CHCHO [1.5]; Name: 2-methylpropanal [1.5].	[3]

Question 4: Geometric Stereoisomerism in Unsaturated Dicarboxylic Acids: Maleic vs Fumaric Acid [10 Marks]

Part	Marking Points & Expected Observations	Marks
(a)	M (cis / Z): both -COOH groups on the same side of the C=C double bond [2]; N (trans / E): -COOH groups on opposite sides of the C=C double bond [2].	[4]
(b)	In the Z-isomer (M), the two -COOH groups are in close spatial proximity on the same side of the double bond [2]; elimination of H ₂ O occurs easily to form a stable 5-membered cyclic ring [1]; in the E-isomer (N), the -COOH groups are locked on opposite sides and cannot reach each other without breaking the C=C pi bond [1].	[4]
(c)	Fumaric acid (trans) has a higher melting point [1]; linear trans molecules pack more tightly and symmetrically into the crystal lattice, leading to stronger intermolecular forces [1].	[2]

Question 5: Identification and 3D Representation of Chiral Centres in Practical Unknowns [10 Marks]

Part	Marking Points & Expected Observations	Marks
(a)	Butan-2-ol, CH ₃ CH(OH)CH ₂ CH ₃ [1]; possesses a chiral centre (asymmetric carbon atom, C2) [1].	[2]
(b)	-H, -OH, -CH ₃ (methyl), and -CH ₂ CH ₃ (ethyl) [2].	[2]

(c)	Correct 3D tetrahedral geometry with one wedge, one dashed line, and two solid in-plane lines [2]; mirror plane drawn with non-superimposable mirror images [2].	[4]
(d)	One enantiomer rotates the plane of polarized light clockwise (+ / dextrorotatory); the other rotates it by the exact same angle counterclockwise (- / levorotatory) [2].	[2]

Question 6: Classification of Alcohols using Lucas Reagent (Conc. HCl and Anhydrous ZnCl₂) [10 Marks]

Part	Marking Points & Expected Observations	Marks
(a)	(i) Primary (1°): Solution remains clear at room temperature (no reaction for hours) [1]; (ii) Secondary (2°): Solution turns cloudy after 3 to 5 minutes [1.5]; (iii) Tertiary (3°): Solution turns cloudy immediately within seconds, forming two distinct immiscible layers [1.5].	[4]
(b)	Insoluble chloroalkane (alkyl chloride) [2].	[2]
(c)	The reaction proceeds via an S _N 1 mechanism involving a carbocation intermediate [1]; tertiary carbocations are stabilized by the positive inductive (+I) effect of three electron-releasing alkyl groups, forming almost instantly [1.5]; primary carbocations are extremely unstable, so 1° alcohols cannot react via S _N 1 at room temperature [1.5].	[4]

Question 7: Deductive Identification of an Unknown Halogenoalkane by Hydrolysis Rate and Density [10 Marks]

Part	Marking Points & Expected Observations	Marks
(a)	Bromine (forms cream precipitate of AgBr soluble in concentrated NH ₃) [2].	[2]
(b)	1. 1-bromopropane: CH ₃ CH ₂ CH ₂ Br [2]; 2. 2-bromopropane: CH ₃ CH(Br)CH ₃ [2].	[4]
(c)	1-bromopropane [1]; hydrolysis gives propan-1-ol, which does not contain the CH ₃ -CH(OH)- group and thus gives a negative iodoform test (whereas 2-bromopropane would yield propan-2-ol, giving a positive iodoform test) [1].	[2]
(d)	Ag ⁺ (aq) + Br ⁻ (aq) → AgBr(s) [2].	[2]

Question 8: Diagnostic Deductions of Nitrogen-Containing Organic Compounds: Amines vs Amides [10 Marks]

Part	Marking Points & Expected Observations	Marks
(a)	Primary amine (or secondary amine / amino group -NH ₂) [2].	[2]
(b)	Alkylammonium chloride salt, R-NH ₃ ⁺ Cl ⁻ (e.g. CH ₃ CH ₂ NH ₃ Cl) [2].	[2]
(c)	The nitrogen atom possesses a non-bonding lone pair of electrons [1.5]; it can accept a proton (H ⁺) by forming a dative covalent bond: R-NH ₂ + H ⁺ → R-NH ₃ ⁺ [1.5].	[3]
(d)	CH ₃ CH ₂ NH ₂ + CH ₃ COCl → CH ₃ CONHCH ₂ CH ₃ + HCl (forms N-ethylethanamide) [3].	[3]

Question 9: Deductive Structure Determination Combining Elemental Analysis and Titrimetric Mr [10 Marks]

Part	Marking Points & Expected Observations	Marks
(a)	Moles C = 40.0 / 12.0 = 3.33; Moles H = 6.7 / 1.0 = 6.7; Moles O = 53.3 / 16.0 = 3.33 [1.5]; Ratio = 1:2:1 → Empirical formula = CH ₂ O [1.5].	[3]
(b)	Moles NaOH = 0.200 x 0.0500 = 0.0100 mol [1]; Moles acid = 0.0100 mol [1]; Mr = 0.900 / 0.0100 = 90.0 g mol ⁻¹ [1].	[3]
(c)	Empirical mass CH ₂ O = 12 + 2 + 16 = 30.0; Multiple = 90.0 / 30.0 = 3; Molecular formula = C ₃ H ₆ O ₃ [2].	[2]
(d)	CH ₃ -CH(OH)-COOH [1]; 2-hydroxypropanoic acid (lactic acid) [1].	[2]

Question 10: Comprehensive Laboratory Decision Tree for the Identification of Unknown Organic Liquids [10 Marks]

Part	Marking Points & Expected Observations	Marks
(a)	Add 2,4-dinitrophenylhydrazine (2,4-DNPH / Brady's reagent): bright orange-yellow precipitate confirms a carbonyl (aldehyde or ketone) [2].	[2]

(b)	Tollens' reagent warmed in a water bath: silver mirror confirms aldehyde; clear solution confirms ketone [2] (or Fehling's solution).	[2]
(c)	Add solid sodium hydrogencarbonate (NaHCO_3): rapid effervescence of CO_2 confirms carboxylic acid [1.5]; if negative, test with dry sodium metal: effervescence of H_2 confirms an alcohol [1.5].	[3]
(d)	Warm with acidified potassium dichromate(VI): color change from orange to green confirms 1° or 2° alcohol; remaining orange confirms 3° alcohol [3].	[3]